

Arithmetic Progressions

What an arithmetic progression is, how to find its n th term and how to add up the first n terms, all from NCERT with worked examples and practice questions.

Introduction

Look at these lists of numbers:

1, 3, 5, 7, 9,... and 10, 7, 4, 1, -2,...

In the first list, each number is 2 more than the one before it. In the second list, each number is 3 less than the one before it. Whenever a list follows this kind of steady step, the same fixed number added each time, it is called an **arithmetic progression**. This chapter teaches you how to recognise such lists, find any term without writing out the whole list, and add up the first n terms quickly.

What is an Arithmetic Progression?

Arithmetic Progression (AP)

An AP is a list of numbers in which each term is obtained by adding a fixed number to the preceding term, except the first term. This fixed number is called the **common difference** and is denoted by d .

$a, a + d, a + 2d, a + 3d, \dots$

where a is the first term (often written a_1).

The common difference can be positive, negative or even zero:

- if $d > 0$, the AP increases (for example, 2, 5, 8, 11,... where $d = 3$);
- if $d < 0$, the AP decreases (for example, 10, 7, 4, 1,... where $d = -3$);
- if $d = 0$, every term is the same (for example, 4, 4, 4, 4,...).

To check whether a given list is an AP, see if the difference between any two consecutive terms is the same throughout.

Worked Example

Check whether the list 3, 8, 13, 18,... is an AP. If it is, write the next two terms.

$$a_2 - a_1 = 8 - 3 = 5; \quad a_3 - a_2 = 13 - 8 = 5; \quad a_4 - a_3 = 18 - 13 = 5.$$

The difference is the same each time ($d = 5$), so this **is an AP**. The next two terms are $18 + 5 = 23$ and $23 + 5 = 28$.

A useful test that follows from the definition: three numbers a , b and c are in AP if and only if $2b = a + c$. That is, the middle term is the average of the other two.

The n th Term of an AP

Writing out a long AP term by term is slow. The n th term formula lets you jump straight to any term.

n th Term (General Term) of an AP

If the first term is a and the common difference is d , then the n th term is

$$a_n = a + (n - 1)d$$

This is also called the **general term** of the AP. If the last term of a finite AP is denoted by l , then $l = a + (n - 1)d$.

The formula makes sense if you think about it: to get from the 1st term to the 2nd you add d once, to the 3rd you add d twice, and so to the n th you add d exactly $(n - 1)$ times.

Worked Example

Find the 20th term of the AP 7, 13, 19,...

Here $a = 7$ and $d = 13 - 7 = 6$.

$$a_{20} = a + (20 - 1)d = 7 + 19 \times 6 = 7 + 114 = 121$$

So the **20th term is 121**.

Worked Example

Which term of the AP 3, 8, 13, 18,... is 78?

Here $a = 3$, $d = 5$, and we want $a_n = 78$.

$$a + (n - 1)d = 78 \implies 3 + (n - 1) \times 5 = 78$$

$$(n - 1) \times 5 = 75 \implies n - 1 = 15 \implies n = 16$$

So **78 is the 16th term** of this AP.

These two types of question, "find a_n given a and d " and "find n given a_n ", are the most common in exams. Another common type is: "How many terms are there?" which is just finding n when the last term l is known.

Sum of the First n Terms

Adding many terms of an AP one by one is tedious. There is a short formula that does it in one step. It comes from a nice trick: write the sum forwards and backwards and add.

Sum of the First n Terms of an AP

If the first term is a , the common difference is d and the last term is $l = a_n$, then

$$S_n = n/2 [2a + (n - 1)d]$$

and, when the last term l is known,

$$S_n = n/2 (a + l)$$

Both forms are the same, the second is just a shorter version of the first when you already know l .

Why does it work? Pair the first term with the last, the second with the second-last, and so on. Each pair adds to the same number ($a + l$), and there are $n/2$ such pairs. That is exactly what $S_n = n/2 (a + l)$ says.

Note that S_n here means the sum of the first n terms. The n th term itself can also be recovered from sums: $a_n = S_n - S_{n-1}$ (for $n > 1$).

Worked Example

Find the sum of the first 22 terms of the AP 8, 3, -2,...

Here $a = 8$, $d = 3 - 8 = -5$, $n = 22$.

$$S_{22} = 22/2 [2(8) + (22 - 1)(-5)] = 11 [16 - 105] = 11 \times (-89) = -979$$

So the **sum of the first 22 terms is -979**.

Worked Examples

Worked Example

Find the sum of the first 15 terms of the AP whose n th term is $a_n = 2 + 3n$.

This is an AP because a_n is linear in n . We can find a_1 and a_2 to get a and d , or use the formula directly. First find the first and last terms:

$$a_1 = 2 + 3(1) = 5, \quad a_{15} = 2 + 3(15) = 47$$

Now use $S_n = n/2 (a + l)$:

$$S_{15} = 15/2 (5 + 47) = 15/2 \times 52 = 15 \times 26 = 390$$

So the **sum is 390**. You can also verify that $d = a_2 - a_1 = 8 - 5 = 3$, and $S_{15} = 15/2 [2(5) + 14(3)] = 15/2 \times 52 = 390$.

Worked Example

A sum of ₹ 1000 is invested at 8% simple interest per year. What will the amount be after 4 years?

Simple interest each year is constant: 8% of ₹ 1000 = ₹ 80. So the amounts at the end of year 1, 2, 3, 4 form an AP: 1080, 1160, 1240, 1320,... where each term is the previous plus ₹ 80.

More directly, the amount after n years is $1000 + 80n$, which is an AP with $a = 1080$ and $d = 80$. After 4 years: $1000 + 80 \times 4 = \text{₹ } 1320$.

This shows how APs naturally model situations with a constant yearly increase, such as simple interest or fixed savings.

Summary

- **AP definition:** a list where each term (after the first) is obtained by adding a fixed number d to the preceding term. General form: $a, a + d, a + 2d, a + 3d, \dots$
- **Common difference:** $d = a_2 - a_1 = a_3 - a_2 = \dots$ It can be positive, negative or zero. A list is an AP only if this difference is constant.
- **n th term (general term):** $a_n = a + (n - 1)d$. Use it to find any term or to find n when a_n is known. Three numbers a, b, c are in AP iff $2b = a + c$.
- **Sum of first n terms:** $S_n = n/2 [2a + (n - 1)d] = n/2 (a + l)$, where $l = a_n$ is the last term. Also, $a_n = S_n - S_{n-1}$ for $n > 1$.
- **Finite vs infinite AP:** a finite AP has a last term l ; an infinite AP goes on forever. Both use the same formulas for a_n and S_n .

Practice Questions

1. Check whether the list 2, 4, 8, 16,... is an AP. Give a reason.

2. Find the 30th term of the AP 10, 7, 4,...

3. The 5th term of an AP is 16 and the 12th term is 37. Find the first term and the common difference.

4. How many terms of the AP 9, 17, 25,... must be taken to give a sum of 636?

5. The sum of the first n terms of an AP is $S_n = 3n^2 - 4n$. Find the n th term and the common difference.

6. A manufacturer produces 50 articles in the first year and increases production by 10 articles every year. How many articles will be produced in the 15th year, and what is the total production in 15 years?